

Complete Patient-Level Leakage in a Widely Used Brain Tumor MRI Benchmark

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Abstract

Brain tumor classification from MRI is a popular deep learning benchmark, and reported accuracies on the widely used four-class aggregated dataset routinely exceed 98%. We audit that dataset and show the figure is substantially an artifact of how it is split. Recovering patient identifiers by content-matching the tumor images back to their figshare source, we find that 206 of 233 patients have slices in both the training and test folders; of the 906 tumor test images, 761 (84%) could be traced, and every one of them belongs to a patient the model trains on. The dataset also contains 216 test images (16.5%) that are pixel-identical to a training image, and holds only 6,597 distinct images among its 7,023 files. Removing the duplicate images changes little (99.22% to 99.42%; the 95% interval on the difference is $[-0.03, +0.43]$ points), because it leaves every repeated patient in place. Removing the patient overlap changes a great deal: under five-fold patient-disjoint cross-validation accuracy falls to 95.19%, and to 93.68% on the tumor classes alone. In a paired comparison on the 761 test images traceable to a patient, scored under both regimes, accuracy is 99.12% when the patient appeared in training and 93.60% when it did not, a 7.3-fold increase in error rate. We also find that partition choice contributes 2.6 times the standard deviation that random seeding does, so single-split accuracies on this benchmark, including our own earlier ones, carry far wider error bars than usually reported. We release the recovered patient map and a fixed patient-disjoint partition, and we report two corrections to our own earlier analyses: a provenance result that reversed once its positive control was made adequate, and a significance test that was invalid under clustering.

Keywords: data leakage; dataset audit; patient-level splitting; brain tumor MRI; benchmark contamination; reproducibility; transfer learning; medical image classification

1 Introduction

A brain tumor is an abnormal growth of cells inside the skull, and depending on type and location, it can be life-threatening. Tumors are classified as primary, meaning they originate in the brain, or secondary, meaning they spread from cancer

elsewhere in the body. Among primary tumors, gliomas and meningiomas are the most common, with gliomas generally being more aggressive. Because symptoms depend heavily on tumor size and location, early and accurate detection matters a great deal. Diagnosis has traditionally relied on radiologists reading MRI scans by hand, a process that takes time and is subject to disagreement between readers. Given how much a delayed or missed diagnosis can affect a patient’s outcome, there is real value in tools that can support this process.

Deep learning, and CNNs in particular, has changed how image analysis is done by learning hierarchical features directly from pixels instead of relying on hand-crafted rules. This has made it possible to segment and classify medical images without manually engineering features for every task. When labeled data is limited, which is common in medical imaging, transfer learning helps close the gap: a CNN pretrained on a large natural-image dataset like ImageNet is fine-tuned on the smaller medical dataset instead of being trained from nothing. Tajbakhsh et al. (2016) showed that fine-tuned pretrained models frequently outperform models trained from scratch on medical imaging tasks. CNNs have since been applied widely to both segmentation and classification of brain MRI. Havaei et al. (2017), for example, combined local and global image context in a single CNN to achieve strong glioblastoma segmentation results. On the classification side, several recent studies report high accuracy with transfer learning: Disci et al. (2025) fine-tuned architectures such as Xception, MobileNetV2, InceptionV3, ResNet50, VGG16, and DenseNet121 on the same four-class dataset used here and reported strong classification accuracy, while Babu Vimala et al. (2023) used a hybrid deep learning approach to reach high test accuracy distinguishing glioma, meningioma, and pituitary tumors.

The contribution of this paper is not a new architecture. It is an audit of a widely used benchmark, and a re-evaluation of a standard baseline on a split that is actually valid. Specifically:

- We recover the dataset’s provenance by content matching against the figshare collection of Cheng et al. (2015), and in doing so recover a patient identifier for 91.3% of its tumor images (Section 3.1). We describe the calibrated control this required, and the false negative we reached

without it, because the methodological lesson generalizes to any such trace.

- Using those identifiers, we show that the released split has **complete patient-level leakage**: 100% of test images with a recoverable patient come from a patient also present in training (Section 3.3). No evaluation on this split measures generalization to an unseen patient.
- We document duplicate contamination as a separate, smaller defect: 16.5% of test images are pixel-identical to a training image (Section 3.2), and the dataset’s 7,023 files contain 6,597 distinct images (Section 3.4).
- We re-evaluate the same pipeline under five-fold patient-disjoint cross-validation built from the recovered identifiers (Section 6.8). Accuracy falls from 99.22% to 95.19%, and to 93.68% on the tumor classes alone; a paired comparison on the same 761 traceable images gives a 7.3-fold increase in error rate. This is the paper’s main experimental result.
- We show that deduplication alone does not change accuracy, and explain why: it removes repeated images while leaving every repeated patient in place (Sections 6.7 and 3.3). We also show the duplicated images are misclassified twice as often as the rest, so they were never being memorized.
- We report the baseline completely rather than as a headline number: per-class metrics, confusion matrix, ROC-AUC, ten-seed variance, a five-seed-per-cell ablation, and temperature-scaled calibration (Sections 6 to 6.6).

A note on how to read the accuracy figures in this paper. Where we quote a number near 99%, it is a measurement on the released split, which we show does not measure generalization to an unseen patient. Those figures appear so that others can reproduce them and so that the split comparisons have a baseline, not as a claim about performance. Where we want to state what this pipeline actually achieves, we use the patient-disjoint figure of Section 6.8, and neither number is a claim of clinical readiness.

2 Related Work

CNNs have performed well across a range of medical imaging tasks, including tumor segmentation and detection. Havaei et al. (2017) introduced a fully automatic brain tumor segmentation network that improved both speed and accuracy over earlier methods by combining local and more global contextual features. Pereira et al. (2016) took a related approach, building a deep CNN for glioma segmentation using small convolutional kernels stacked in depth. These segmentation studies motivate our use of CNNs for classification, since the same hierarchical features that trace tumor boundaries can plausibly distinguish between tumor types.

Transfer learning is a natural fit for classification problems with limited labeled data. Tajbakhsh et al. (2016) studied medical image analysis tasks spanning radiology and cardiology

and found that CNNs pretrained on ImageNet and then fine-tuned generally outperform networks trained from scratch, and that fine-tuned networks are also more robust when training data is scarce. MRI scans look nothing like ImageNet’s natural photographs, but this line of work is exactly why we still start from an ImageNet-pretrained ResNet-18: fine-tuning lets the network adapt its low-level filters to MRI texture while keeping the general-purpose pattern detectors learned from a much larger dataset.

A large share of this literature builds on the T1-weighted contrast-enhanced brain tumor dataset released by Cheng et al. (2015) on Figshare, which covers the same three tumor types (glioma, meningioma, pituitary) used here and has become a de facto benchmark; several of the public four-class collections used in more recent work, including datasets used by studies discussed below, trace part of their images back to it. Swati et al. (2019) were among the first to apply transfer learning to this dataset directly, using a block-wise fine-tuning strategy on a pretrained VGG19 rather than fine-tuning the whole network at once, which let them study how much of a pretrained CNN actually needs to change for a new imaging domain. Deepak and Ameer (2019) took a feature-extraction approach instead, using a pretrained GoogLeNet purely to produce features and training a separate classifier on top, and reported a mean classification accuracy of 98% across glioma, meningioma, and pituitary tumors. Khan et al. (2020) trained a CNN classifier on a smaller, independently annotated brain MRI collection, which is a useful reminder that results on the large aggregated datasets used elsewhere in this section do not automatically transfer to differently sourced data.

Several more recent studies broaden the comparison to more backbones or add a second technique on top of transfer learning. Disci et al. (2025) benchmarked six pretrained architectures, including Xception, ResNet50, and DenseNet121, on the same public four-class brain MRI dataset used in this paper and reported strong classification accuracy through transfer learning and fine-tuning. Babu Vimala et al. (2023) applied a hybrid deep learning approach combining EfficientNet-based transfer learning with additional classifiers on the Figshare brain tumor dataset and reported high accuracy across glioma, meningioma, and pituitary tumor classes. Srinivas et al. (2022) compared VGG-16, ResNet-50, and Inception-v3 directly against each other under the same transfer learning setup and found meaningful accuracy differences between backbones on the same data, which is a useful caution against assuming any one pretrained network is the right default choice. Krishnapriya and Karuna (2023) ran a similar head-to-head comparison across a wider set of pretrained models. Anaya-Isaza et al. (2023) went further and compared standard transfer learning against data augmentation and a cross-transformer architecture on the same classification and detection tasks, which starts to probe whether the gains in this literature come from the backbone, the augmentation, or the training recipe. Dutta et al. (2022) is a smaller-scale study in the same vein, applying transfer learning-based deep networks to the multi-class version of the problem. Other work looks at combining models rather than picking one: Chen et al. (2024) fused deep fea-

tures from multiple pretrained CNNs to improve the robustness of multi-type brain tumor classification, reporting accuracy above 95% on their evaluation set; Sankar et al. (2024) built an ensemble of deep learning models specifically for tumor *grade* classification, a related but distinct problem from the type classification addressed here; Gasmi et al. (2024) combined several deep models with a weight-selection technique to merge their predictions; and Alkharaan et al. (2026) paired an EfficientNet classifier with a U-Net++ segmentation branch so that classification and tumor localization are produced together rather than as separate steps.

Taken together, these results suggest that pretrained CNNs, given careful fine-tuning and reasonable data augmentation, can classify brain tumors with high accuracy across a wide range of backbones and training recipes. What almost none of them examine is the split they are measured on. Of the studies surveyed above, none report checking their evaluation split for duplicated images, and none report verifying that the patients in their test set are absent from their training set. We make that observation about their published descriptions, not about what they may have done and not reported.

2.1 Leakage and Data Quality in Machine Learning Benchmarks

The failure mode this paper documents is not new, and it would be misleading to present it as such. Kapoor and Narayanan (2023) survey machine-learning-based science across 17 fields and find leakage affecting 294 papers, offering a taxonomy in which our case is an instance of the category where the test set is not independent of the training set. Roberts et al. (2021) reviewed 2,212 papers on machine learning for COVID-19 imaging and found none of the 62 they examined in depth to be of clinical use, with reliance on assembled public datasets containing duplicated and non-independent images among the recurring causes. That is the closest precedent for what we report here: a sizable literature accumulating on a benchmark whose construction had not been checked.

In medical imaging specifically, leakage from splitting at the slice level rather than the subject level is well documented. Yagis et al. (2021) quantified precisely this effect for 2D CNNs on brain MRI, and found that slice-level splitting inflated apparent accuracy by 29 to 55 percentage points across four datasets. Wen et al. (2020) reached a similar conclusion for CNN-based Alzheimer’s classification, showing that a large share of reported performance in that literature was attributable to subject-level leakage rather than to the methods being compared. Our results are an instance of the phenomenon these papers describe, on a dataset where, to our knowledge, it had not previously been measured, and the size of the effect we observe (Section 6.8) is smaller than theirs but of the same character.

Duplicate contamination between splits has also been found in general vision benchmarks. Barz and Denzler (2020) showed that 3.3% of the CIFAR-10 test set and 10% of CIFAR-100 have duplicates in the corresponding training set, and released corrected versions. The 16.5% we measure here (Sec-

tion 3.2) is larger than either. Oakden-Rayner (2020) audited two widely used public medical imaging datasets by visual inspection and found label quality substantially worse than the accompanying documentation claimed, which is a reminder that a dataset’s popularity is not evidence that anyone has checked it.

Against that background, we are careful about what we claim as new. Detecting leakage is not new; quantifying its cost is not new; recommending subject-level splits is not new, and it is already standard practice in datasets such as CheXpert and MIMIC-CXR, which ship patient-level splits by construction. What has not been done for this particular collection, and what makes the measurement possible at all, is the recovery of patient identity for a dataset that had discarded it. The aggregated release contains no patient identifiers, so the leakage was not merely unmeasured but appeared unmeasurable. Sections 3.1 and 3.3 recover those identifiers by content-matching the images back to their source collection, which is what converts a suspicion into a measurement and a corrected split.

3 Dataset and Audit

We use the brain MRI dataset released by Ghaffar (2024) on Mendeley Data in 2024, which contains 7,023 T1-weighted, contrast-enhanced MRI images spanning four classes: glioma, meningioma, pituitary tumor, and no tumor. The same collection is distributed on Kaggle (Nickparvar, 2021), which documents its composition as a combination of the figshare dataset of Cheng et al. (2015), the SARTAJ collection, and the Br35H set of normal scans; the Mendeley entry appears to mirror it. The images are JPEG-formatted axial and coronal slices, pre-split by the dataset authors into training and testing folders. Our pipeline follows that split as published, treating the training folder as the only source of trainable data and never touching the testing folder until final evaluation, so no test image influences the weights or the choice of checkpoint through our procedure. As Sections 3.2 and 3.3 show, that discipline is not sufficient: the published split itself places the same images, and all of the same patients, on both sides of the boundary before we see it.

Table 1 gives the exact composition of both splits as counted directly from disk. The classes are reasonably balanced, with `notumor` the largest at 1,595 training images and `glioma` the smallest at 1,321. Figure 1 shows example scans from each class, and Figure 2 plots the same counts visually. Both figures are generated by `visualizer.py` from whatever copy of the data is on disk rather than hardcoded, so they always describe the data actually being trained on.

Table 1: Dataset composition, counted from disk. The training split is further divided 80/20 into 4,569 training and 1,143 validation images.

Class	Training	Testing	Total
glioma	1,321	300	1,621
meningioma	1,339	306	1,645
notumor	1,595	405	2,000
pituitary	1,457	300	1,757
Total	5,712	1,311	7,023

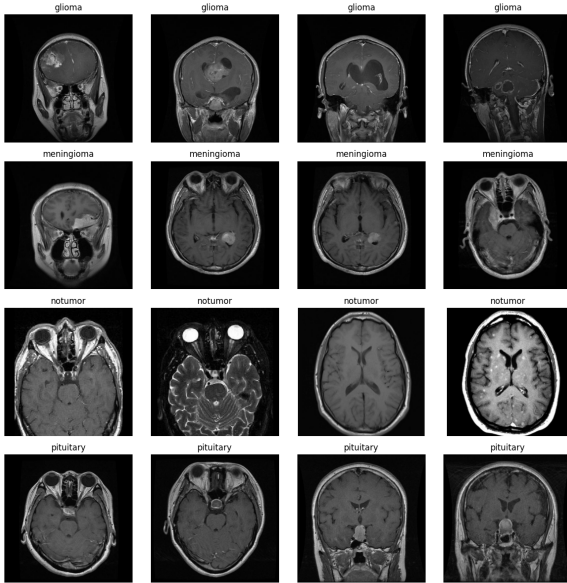


Figure 1: Sample MRI scans from each of the four classes.

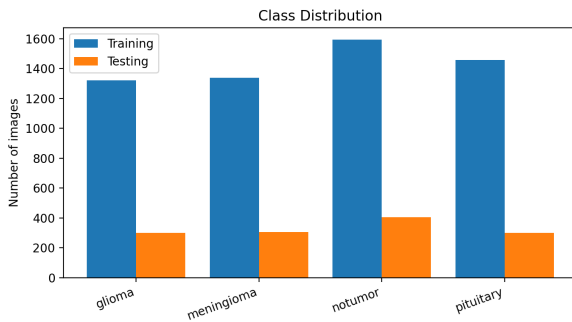


Figure 2: Number of images per class in the training and testing splits, generated by `visualizer.py`.

To match ImageNet-pretrained models such as ResNet-18, every image is resized to 224×224 and grayscale scans are copied across three channels to look like RGB input. Pixel values are then normalized using ImageNet’s mean and standard deviation so the pretrained convolutional filters and batch-normalization statistics behave as intended. This dataset is used widely for benchmarking brain tumor classification

(Disci et al., 2025; Babu Vimala et al., 2023), and its open access and balanced classes make it a practical choice for reproducible research.

That popularity is why the rest of this section is worth the space. Before reporting any accuracy on this benchmark, we audited it. Three findings follow, and anyone using this dataset should know all of them: its train and test splits overlap at the level of individual images; they overlap completely at the level of patients; and the class with no recoverable patient structure is substantially redundant against itself. Recovering the provenance, which turned out to be exactly as documented, is what made the second of those measurable.

3.1 Provenance

Aggregated brain MRI collections of this shape are widely described as combinations of three public sources: the T1-weighted contrast-enhanced dataset of Cheng et al. (2015), the SARTAJ collection, and the Br35H set of normal scans. If the tumor classes here derive from Cheng et al. (2015), that would matter a great deal for evaluation, because that release stores a patient identifier with each of its 3,064 slices and contains multiple slices per patient. A split made by shuffling slices rather than by patient would then place the same patient on both sides of the train/test boundary, and the test set would no longer measure generalization to unseen patients.

Establishing this by content matching turned out to be harder than it looks, and we describe the failure as well as the result because the failure is instructive.

Our first attempt downloaded all 3,064 figshare slices (233 patients), rendered each to 8-bit using exactly the conversion the dataset’s own README documents, and matched all 5,023 tumor-class images against all 3,064 slices under three measures: cosine similarity of ImageNet-pretrained ResNet-18 features, correlation after per-image histogram equalization, and intersection-over-union of binarized brain masks. Nothing matched. The best equalized correlation was 0.872 and the best embedding cosine 0.963, against a median of 0.838 for randomly chosen unrelated pairs. On that basis we initially concluded that the attribution to Cheng et al. (2015) could not be confirmed.

That conclusion was wrong, and the reason is worth stating plainly. Our positive control was a pair of byte-identical images, which establishes only that the measures return 1.0 on identical inputs. It does not establish that they survive the transformations a release pipeline actually applies. The control that matters takes a figshare slice, pushes it through a plausible pipeline, and matches the result back to its own source. When we built that control (`provenance_sensitivity.py`), the measures held up under JPEG re-encoding at quality 85 to 95 and under resizing, both above 0.999, but collapsed under cropping to the brain region: equalized correlation fell to a median of 0.601 and embedding cosine to 0.963. That last figure is exactly the best score we had reported as evidence of non-derivation. Our negative result was not evidence of absence; it was a measurement that could not tell a cropped true match from an unrelated image.

Two corrections fix it. First, crop-normalize both sides, taking the brain bounding box and resizing to a common square, so a difference in framing cannot suppress the similarity. Second, search all eight dihedral orientations of each image, which we had not done: the aggregated images are rotated 90 degrees relative to figshare, and a substantial minority are also mirrored. Under this procedure the same control recovers a genuine transformed match to a median cosine of 0.9988 with a minimum of 0.9884, which calibrates an acceptance threshold rather than leaving it to guesswork.

The corrected search reverses the finding entirely. Table 2 summarizes it. 4,586 of the 5,023 tumor images, 91.3%, match a figshare slice at cosine 0.98 or above, and all 233 figshare patients are represented. Of the accepted matches, 2,956 are at `rot90` and 1,630 at `mirror_rot90`; not one matches at the identity orientation, which is precisely why the original search found nothing. Three images labelled pituitary in this dataset are labelled meningioma in figshare, a small labelling error introduced somewhere in the aggregation.

Table 2: Crop-normalized, orientation-searched matching against figshare. The control is a genuine match pushed through a crop/resize/JPEG pipeline, and calibrates what score a true derivation achieves.

Quantity	Value
Control: genuine match, median cosine	0.9988
Control: genuine match, minimum cosine	0.9884
Acceptance threshold	0.98
Tumor images matched	4,586 / 5,023 (91.3%)
figshare patients recovered	233 / 233
Matched at <code>rot90</code>	2,956
Matched at <code>mirror_rot90</code>	1,630
Matched at identity orientation	0
Label disagreements	3

The documented provenance is therefore correct, and the practical consequence is the opposite of what we first reported: patient identifiers *are* recoverable for the large majority of tumor images, which makes it possible to ask the question the next section asks. We note two limits. We tested the figshare component only, and did not attempt to verify the SARTAJ or Br35H attributions; the `notumor` class in particular has no recoverable patient structure. And 437 tumor images (8.7%) did not match at threshold, which may reflect the SARTAJ component, further transformation, or matching failures we have not diagnosed.

3.2 Duplicate Contamination

The second finding needs no thresholds to state. We hashed every image in both splits, first by raw file bytes and then by decoded pixel content at a fixed resolution, so that an image re-encoded at a different JPEG quality still collides with its original. Table 3 reports the result.

Table 3: Duplicate contamination in the released split. Percentages are of the 1,311-image test set.

Category	Count	% of test
Byte-identical test/train files	103	7.9%
Pixel-identical test/train images	216	16.5%
Duplicates within training split	207	
Duplicates within test split	30	2.3%

216 of the 1,311 test images, 16.5% of the test set, are pixel-for-pixel the same image as something the model trains on, and 103 of those are identical at the level of the file itself. We verified a sample byte for byte. Figure 3 shows examples: these are not similar patients or adjacent slices, they are the same scan appearing twice.

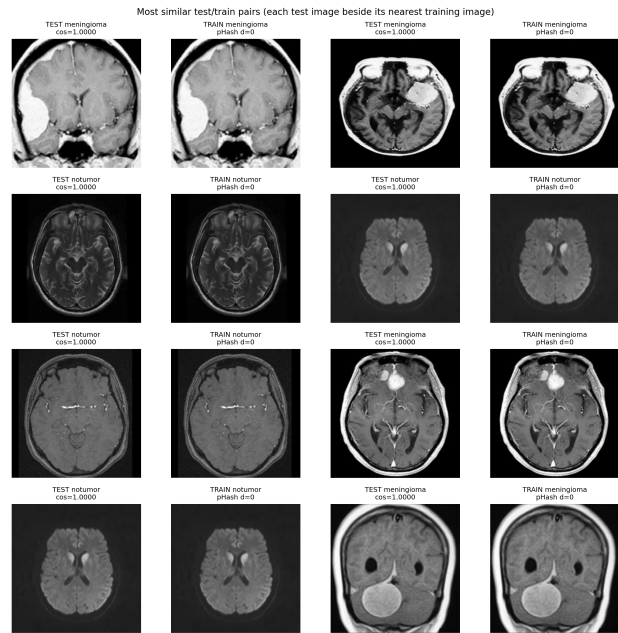


Figure 3: Test images (left of each pair) beside their nearest training image. Every pair shown is pixel-identical. Generated by `audit_leakage.py`.

The contamination is very unevenly distributed across classes, which turns out to matter for interpreting the results later. Table 4 breaks it down. The `notumor` class is the worst affected in absolute terms, and `glioma` is essentially untouched: not one glioma test image duplicates a training image.

Table 4: Pixel-identical duplicates by class. Test column counts test images duplicating some training image.

Class	Test dupes	% of class	Within train
glioma	0	0.0%	0
meningioma	106	34.6%	6
notumor	105	25.9%	189
pituitary	5	1.7%	12

Relaxing to near-duplicates rather than exact ones, 200 further test images sit at cosine similarity 0.99 or above to a training image without being pixel-identical, 197 of them in `notumor`. We treat that number with more caution than the exact counts, because brain MRI slices are generically similar to one another: in our data the nearest different-class training image already sits at roughly 0.90 cosine, so the metric is not sharply discriminative and the threshold is a judgment call. The exact-duplicate counts require no such judgment, and they are what we rely on.

The redundancy of the `notumor` class deserves a separate mention: it contributes 189 of the 207 within-train duplicates and 197 of the 200 near-duplicates. Section 3.4 quantifies this precisely by asking a different question of the same data: not how much overlaps across the train/test boundary, but how much of the dataset is distinct from itself at all.

3.3 Patient-Level Leakage

Duplicate contamination is the visible symptom. The patient identifiers recovered in Section 3.1 let us measure the underlying problem, and it is considerably worse.

Each figshare patient contributes multiple slices, so a split made by shuffling slices rather than by patient puts the same brain, often the same tumor a few millimetres further along, on both sides of the evaluation boundary. Grouping the 4,586 matched images by their recovered patient identifier gives the result in Table 5. Of the 233 patients, 206 (88.4%) have slices in both the training and test folders. Every single matched test image, 761 of 761, belongs to a patient that also appears in training. That holds at 100% for all three tumor classes independently.

Table 5: Patient-level leakage in the released split. “Traced” counts test images for which a figshare patient identifier was recovered; the leakage column is over those images only.

Class	Test images	Traced	Patient in training
glioma	300	287 (96%)	287 (100%)
meningioma	306	178 (58%)	178 (100%)
pituitary	300	296 (99%)	296 (100%)
All tumor	906	761 (84%)	761 (100%)

This is not partial contamination that a careful reader might discount. Among the tumor test images whose patient could be identified, the released test split contains no unseen patients at all. Coverage is uneven across classes, and the caveat matters most for meningioma, where only 58% of test images could be traced: the claim is airtight for the 84% of tumor test images we can account for, and silent about the rest. Section 8 returns to whether the untraced remainder could be materially different. Any accuracy measured on it describes how well a model recognizes additional slices of brains it has already studied, which is a substantially easier task than the one the benchmark is taken to represent, and not the task that matters clinically.

The deduplication of Section 6.7 does not touch this. Removing every pixel-identical test image still leaves 756 of 756 matched test images, and all 206 shared patients, with their patient represented in training; the more aggressive near-duplicate removal gives exactly the same 100%. This is the fact that explains the otherwise puzzling null result in Section 6.7, and we return to it there.

3.4 How Much Distinct Data Is There?

We pooled both splits and counted, per class, the number of files, the number of distinct pixel contents, and the number of near-duplicate clusters at cosine similarity 0.99 (a conservative greedy clustering that never merges two cluster representatives, so it undercounts redundancy if anything). Table 6 reports the result.

Table 6: Redundancy within the full dataset (both splits pooled). “Unique” is exact pixel content; “clusters” additionally collapses near-duplicates at cosine ≥ 0.99 .

Class	Files	Unique	Clusters
glioma	1,621	1,620	1,620
meningioma	1,645	1,531	1,522
notumor	2,000	1,706	1,113
pituitary	1,757	1,740	1,739
Total	7,023	6,597	5,994

The dataset advertises 7,023 images but contains 6,597 distinct ones, and by the more aggressive near-duplicate criterion, roughly 5,994. As with the cross-split contamination, this is almost entirely a `notumor` problem: that class is 44.4% redundant against itself once near-duplicates are collapsed, compared to 0.1% for glioma and roughly 1 to 8% for the other two tumor classes. We were not able to trace this to a specific upstream source; the `notumor` class is commonly attributed to the Br35H collection, but confirming that would require obtaining Br35H directly and repeating the matching procedure of Section 3.1 against it, which we did not do, lacking credentialed access at the time of writing. What Table 6 establishes without needing that trace is that whatever generated the `notumor` class produced far less independent information than its file count suggests, independent of and in addition to the train/test overlap already documented.

4 Methodology

Figure 4 outlines the full pipeline. An input MRI scan is preprocessed to match the format ResNet-18 expects, passed through the pretrained backbone and a newly initialized four-class head, and converted to class probabilities. The resulting predictions feed both the quantitative evaluation and the Grad-CAM interpretability step described in Sections 6 and 7.

Because pretrained CNNs offer a strong balance of representational power and computational cost, we build on

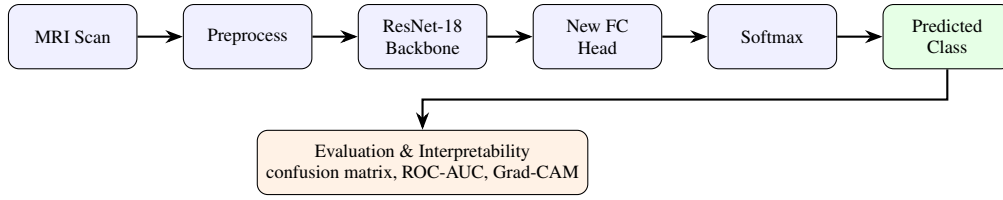


Figure 4: End-to-end pipeline. Preprocessing, the ResNet-18 backbone, and the new classification head are described in Sections 4.1 and 4.2; evaluation and interpretability are described in Sections 6 and 7.

ResNet-18 (He et al., 2016) throughout, though the accompanying code also supports ResNet-34 and EfficientNet-B0 as drop-in alternatives.

4.1 Data Preprocessing

Figure 5 shows the preprocessing chain. All MRI images are resized to 224×224 pixels to match ResNet-18’s expected input. Since the source images are grayscale, the single channel is duplicated three times to imitate an RGB image. Pixel values are then normalized using ImageNet’s mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). These steps keep the pretrained model’s early filters and batch-normalization statistics operating in the input range they were trained on. When augmentation is enabled, a random horizontal flip, a rotation of up to 10 degrees, and brightness and contrast jitter of up to 10% are applied before normalization. None of these transformations change the diagnostic label of the scan, which is what makes them safe to use here.

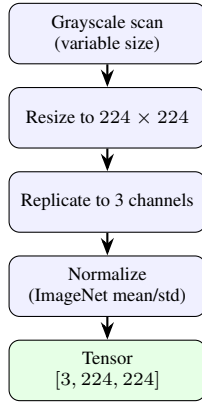


Figure 5: Preprocessing steps applied to every scan before it reaches the network.

4.2 Model Architecture

We use ResNet-18 as implemented in `torchvision.models`, pretrained on ImageNet. Figure 6 shows the layout: the convolutional stem and four residual stages carry over their pretrained weights, and the final fully connected layer, originally sized for 1,000 ImageNet classes, is replaced with a new linear layer with four outputs, one per tumor class. The entire network is fine-tuned rather than just the new head, since full fine-tuning has been shown to

outperform freezing early layers on many medical imaging tasks (Tajbakhsh et al., 2016). The code also supports freezing the backbone (`--freeze_backbone`) for faster experimentation on smaller GPUs.

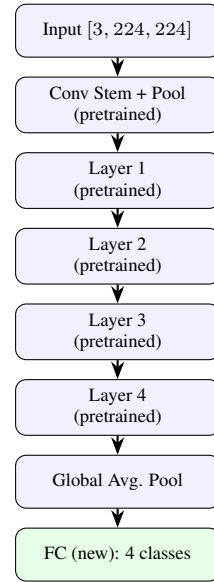


Figure 6: ResNet-18 with its final layer replaced. Blue blocks keep their ImageNet-pretrained weights and are fine-tuned; the green block is initialized from scratch. Layer 4 is also the target layer for Grad-CAM.

4.3 Training Procedure

Training uses cross-entropy loss and the Adam optimizer. The official training folder is split with a stratified 80/20 partition so each class appears in the same proportion in both subsets, giving 4,569 training and 1,143 validation images. A `ReduceLROnPlateau` scheduler halves the learning rate when validation loss stops improving for two epochs, and training stops early if validation loss fails to improve for five consecutive epochs. The checkpoint is saved whenever validation loss reaches a new minimum, so the model that gets evaluated is the best one seen during training and not simply the last one. All random number generators are seeded, and the checkpoint stores the class names and backbone choice alongside the weights, so evaluation and inference never have to guess the class order.

5 Experimental Setup

Table 7 lists every hyperparameter used to produce the results in this paper. Training ran on a single NVIDIA RTX 4070 with PyTorch 2.14 and CUDA 12.6, and each run completed in between roughly 8 and 20 minutes; the same script runs unchanged on CPU, just considerably slower. Evaluation metrics come from `scikit-learn`, and every figure and table in this paper is produced directly by the released scripts.

The main result (Table 8 onward) uses seed 42. Section 6.5 reports ten-seed and five-seed-per-cell ablation runs produced by `train.py` with different `--seed`, `--augment`, and `--freeze_backbone` settings, orchestrated by `run_seed_sweep.py`. Section 6.6 uses `calibrate.py` to fit and evaluate temperature scaling. The audit in Section 3.2 is produced by `audit_leakage.py`, the provenance analysis in Section 3.1 by `match_figshare.py`, and the deduplicated split in Section 6.7 by `build_clean_split.py`. Every per-run metric behind every table is written to JSON by those scripts and included in the repository, so the aggregate numbers can be checked without rerunning anything.

Table 7: Hyperparameters used for the reported run.

Setting	Value
Backbone	ResNet-18 (ImageNet)
Fine-tuning	All layers
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	32
Max epochs	30
Early stopping patience	5
LR scheduler	ReduceLROnPlateau
Scheduler factor / patience	0.5 / 2
Augmentation	Flip, $\pm 10^\circ$, jitter
Random seed	42

6 Results

6.1 Training Dynamics

Figure 7 shows loss and accuracy per epoch. The model converges quickly: validation accuracy passes 98% by the third epoch, which is unsurprising given that the backbone starts from pretrained weights and only has to adapt. After roughly epoch 10 the training loss falls to near zero while validation loss flattens out in the 0.04 to 0.05 range and moves around without trending down. That widening gap is textbook mild overfitting, and it is precisely the situation early stopping exists to handle. Training halted at epoch 21 after five epochs without improvement, and the best checkpoint by validation loss came from epoch 16 (validation loss 0.0376, validation accuracy 99.13%). Peak validation accuracy over the run was 99.21% at epoch 15, slightly higher than at the checkpoint we selected, which is a reminder that selecting on loss and selecting on accuracy do not always agree.

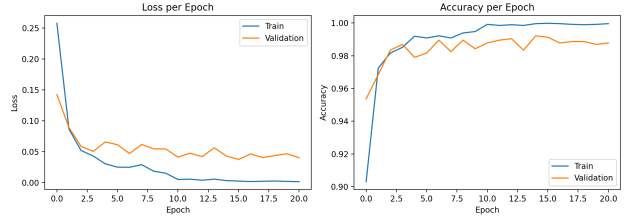


Figure 7: Training and validation loss and accuracy per epoch. Training loss approaches zero while validation loss plateaus, which is what triggers early stopping at epoch 21.

6.2 Test Set Performance

The selected checkpoint was evaluated once on the 1,311 images of the released test folder. Overall accuracy is 99.16%, corresponding to 1,300 correct predictions and 11 errors. Table 8 breaks this down by class.

Table 8: Per-class performance on the 1,311 images of the released test folder.

Class	Prec.	Recall	F1	n
glioma	0.997	0.977	0.987	300
meningioma	0.981	0.994	0.987	306
notumor	0.993	0.998	0.995	405
pituitary	0.997	0.997	0.997	300
Accuracy		0.9916		1,311
Macro avg	0.992	0.991	0.991	1,311
Weighted avg	0.992	0.992	0.992	1,311

Performance is consistent across classes, with F1 scores between 0.987 and 0.997. The weakest class is glioma, whose recall of 0.977 is noticeably below its precision of 0.997. In plain terms, when the model says a scan is a glioma it is almost always right, but it misses about 2% of actual gliomas. Given that gliomas are the more aggressive tumor type, a missed glioma is the most costly error in this matrix, so the asymmetry is worth naming rather than averaging away.

6.3 Confusion Matrix and Error Analysis

Figure 8 shows the normalized confusion matrix, and Table 9 gives the raw counts, which are easier to reason about when the errors are this few.

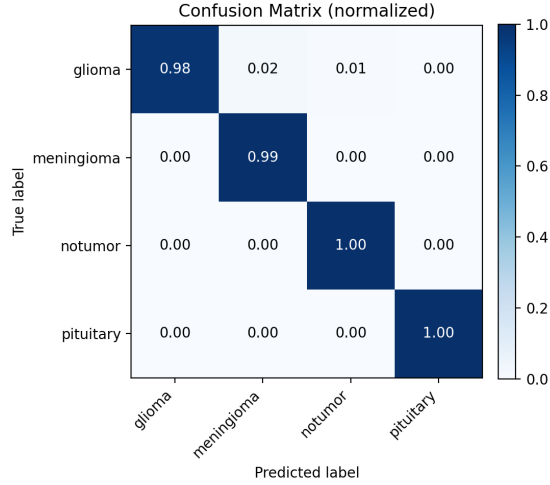


Figure 8: Test-set confusion matrix, normalized by true class.

Table 9: Raw confusion matrix. Rows are true classes, columns are predictions.

True \ Pred	glio.	men.	not.	pit.
glioma	293	5	2	0
meningioma	0	304	1	1
notumor	1	0	404	0
pituitary	0	1	0	299

The errors are not spread evenly. Seven of the 11 come from the glioma class, and five of those are gliomas predicted as meningiomas, making glioma-to-meningioma the single dominant failure mode. This is a clinically sensible confusion: both are common primary tumors, both can present with similar contrast enhancement on a single 2D slice, and distinguishing them often depends on location and volumetric context that a single slice does not carry. The remaining errors are scattered: two gliomas read as no tumor, one meningioma each as no tumor and pituitary, one no-tumor scan read as glioma, and one pituitary as meningioma. The two false negatives where a glioma is called healthy are the most concerning cases in the set, even at only two out of 300.

A detail that a headline accuracy number hides is how confident the model is when it is wrong. Of the 11 errors, four were made with softmax confidence above 0.92, including a glioma called meningioma at 0.964. Only two errors fell below 0.60, which is roughly where a confidence threshold might plausibly route a case for human review. A model that fails loudly and confidently is harder to build a safe workflow around than one whose errors sit near the decision boundary, and this is the clearest evidence in our results that calibration deserves attention before anything like clinical use. Figure 9 shows the misclassified scans themselves.

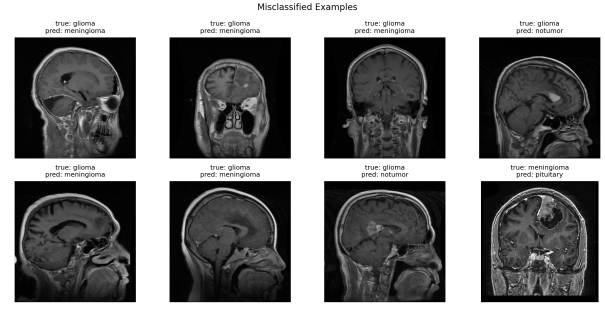


Figure 9: Misclassified test scans with true and predicted labels.

6.4 Ranking Quality

Figure 10 shows one-vs-rest ROC curves. All four classes have an AUC above 0.999 (glioma 0.9999, meningioma 0.9998, notumor 1.0000, pituitary 1.0000), with a macro average of 0.9999. AUC measures how well the model ranks positives above negatives independent of any specific threshold, so these values say the class scores are almost perfectly separable. They should be read alongside, not instead of, the confusion matrix: near-perfect AUC and 11 hard errors at the default threshold are consistent, and together they suggest that a better-chosen operating point could trade some precision for the glioma recall that matters most here.

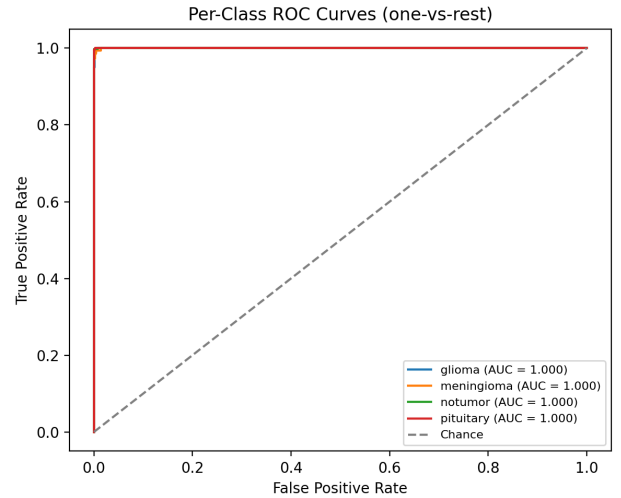


Figure 10: Per-class one-vs-rest ROC curves with AUC.

6.5 Seed Variance and Ablations

A single training run cannot separate a real effect from run-to-run noise, and a single configuration cannot say which parts of the recipe are doing the work. We address both with a 10-seed sweep under the main configuration (augmentation on, full fine-tuning), and a 2×2 ablation over augmentation and fine-tuning strategy, each cell averaged over 5 seeds rather than a single run. Every run used the same hyperparameters as Table 7 and was evaluated once on the same released test folder.

Table 10 reports the 10-seed sweep, generated directly from the released per-run JSON rather than transcribed by hand. One caution about precision: re-scoring identical checkpoints does not always reproduce identical predictions, because GPU inference is not bit-deterministic. Two of these ten runs changed by one image when re-evaluated, which moves the reported accuracy by 0.08 points. That is far smaller than the effects this paper discusses, but it is larger than the last digit of an accuracy quoted to two decimal places, and it is a further reason to treat single-run figures on this benchmark as less precise than they look. Test accuracy ranges from 98.93% to 99.47%, giving a mean of 99.22% with a sample standard deviation of 0.16 points. This confirms what the gap between peak and selected validation accuracy already hinted at: at this accuracy level, a difference of a few tenths of a point between two methods, or between this model and a competing report in Table 16, is within the noise of a single run and should not be read as a real difference without a comparable seed sweep on both sides.

Table 10: Test accuracy across 10 random seeds, same configuration (ResNet-18, augmentation on, full fine-tuning) as the main result.

Seed	Test Acc.	Macro F1	Errors
42	99.16%	0.9913	11
0	99.31%	0.9926	9
1	99.24%	0.9919	10
2	98.93%	0.9891	14
3	99.16%	0.9915	11
4	99.39%	0.9936	8
5	99.47%	0.9946	7
6	99.16%	0.9914	11
7	99.08%	0.9906	12
8	99.31%	0.9927	9
Mean	99.22%	0.9919	10.2
SD	0.16 pt	0.0016	2.0

Table 11 reports the ablation. The effect of freezing the backbone is far larger than anything else studied in this paper: mean test accuracy falls from 99.22% to 83.57% when only the new classification head is trained and the pretrained convolutional layers are left untouched, a drop of 15.65 points ($t = 80.6$, $p < 10^{-6}$ against the same-augmentation full-fine-tuning cell). The same comparison without augmentation gives a 13.67-point drop, equally decisive ($p < 10^{-6}$). Freezing the backbone and only training a linear head is not a reasonable shortcut here, which is a direct, practical confirmation of the claim in Section 4.2, based on Tajbakhsh et al. (2016), that full fine-tuning matters for this kind of transfer.

Augmentation’s effect is much smaller and, once properly averaged over 5 seeds rather than read off a single run, more modest than an earlier single-seed estimate from this project suggested. With full fine-tuning, turning augmentation on improves mean test accuracy from 98.66% to 99.22%, a gain of 0.56 points that is only marginally significant ($t = 2.62$, $p = 0.053$) at this sample size. With the backbone frozen, aug-

mentation’s effect flips sign and becomes clearly significant in the other direction: accuracy falls from 84.99% to 83.57%, a loss of 1.42 points ($t = -6.10$, $p = 0.0004$). A plausible reading is that a frozen backbone’s features were learned on unaugmented natural images, and a linear head sitting on top of those fixed features has limited capacity to absorb the extra input variability that augmentation introduces, so augmentation adds noise without adding anything the fixed features can exploit; with the backbone free to adapt, the same variability instead acts as a mild, marginally useful regularizer. We would not lean hard on the full-fine-tuning augmentation effect given $p = 0.053$, but the fine-tuning effect itself, and augmentation’s harm under freezing, are not in doubt.

Table 11: Ablation over augmentation and fine-tuning strategy. The augment-on/full-fine-tuning cell is the main configuration and reuses all 10 seeds from Table 10; the other three cells are mean $\pm$ SD over 5 seeds each.

Augment	Fine-tuning	Test Acc.	Errors
On	Full	99.22% $\pm$ 0.16	10.2 $\pm$ 2.0
Off	Full	98.66% $\pm$ 0.47	17.6 $\pm$ 6.1
On	Frozen	83.57% $\pm$ 0.42	215.4 $\pm$ 5.5
Off	Frozen	84.99% $\pm$ 0.31	196.8 $\pm$ 4.0

6.6 Calibration

Section 6 noted that several errors are made with high softmax confidence, which is a calibration problem rather than an accuracy problem: a well-calibrated model’s confidence should track its actual probability of being correct, and a model that is confidently wrong is dangerous to route into a workflow that trusts confidence as a proxy for correctness. We fit a single temperature parameter T (Guo et al., 2017) on the validation set by minimizing negative log-likelihood, then divide test-set logits by T before softmax. Temperature scaling cannot change which class is predicted, since dividing all logits by a positive constant does not change their ranking, so it can only reshape confidence, not accuracy or the identity of the 11 test errors.

The fitted temperature was $T = 1.38$, meaning the network’s raw logits were somewhat overconfident and needed softening. Table 12 reports expected calibration error (ECE) and negative log-likelihood (NLL) before and after scaling, on both the validation set the temperature was fit on and the released test folder.

Table 12: Calibration before and after fitting temperature scaling on the validation set. ECE and NLL are lower when better calibrated.

Metric	Validation		Test	
	Before	After	Before	After
ECE	0.0058	0.0034	0.0043	0.0048
NLL	0.0376	0.0339	0.0209	0.0209

The result is more nuanced than a clean before/after improvement, and we think the nuance is worth keeping rather than smoothing over. On the validation set, where T was fit, both ECE and NLL improve, exactly as expected. On the test set, ECE is essentially unchanged (0.0043 to 0.0048, a small increase within noise at only 1,311 examples) and NLL does not move. Test-set calibration was already very good before any correction: an ECE around 0.004 to 0.005 is low, so there was little global miscalibration for temperature scaling to fix, and a temperature fit on 1,143 validation examples does not perfectly transfer to the test set.

What does change, and what we think is the actually useful part of this analysis, is the confidence assigned to the errors that matter most. Of the four test errors originally made with confidence above 0.90 (Section 6), only one remains above 0.90 after temperature scaling; the glioma misclassified as meningioma at 0.964 confidence drops to 0.911, and three other high-confidence errors drop below the 0.90 threshold entirely. In other words, temperature scaling did not move the aggregate calibration metric much on this test set, but it specifically pulled down the confidence of the errors that a confidence-based review threshold would otherwise have let through uninspected. That is a small effect on a single held-out set of 1,311 images, and we would not generalize it beyond this dataset, but it is a concrete, cheap intervention that a deployment could apply for free before deciding what confidence threshold routes a case to a radiologist.

6.7 Does the Contamination Inflate the Result?

Section 3.2 establishes that 16.5% of the test set is duplicated in training. The natural inference is that the accuracy reported above is inflated, and that a clean test set would score materially lower. We tested that inference, and it does not hold.

We built a duplicate-disjoint version of the dataset with `build_clean_split.py`. Every test image that is pixel-identical to a training image is dropped from the test set, duplicate groups within each split collapse to a single representative, and nothing else changes. Dropping from test rather than train keeps the training data intact, so the comparison isolates the effect of cleaning the evaluation set. The result is 5,505 training and 1,092 test images. We then reran the identical pipeline, same hyperparameters, on the clean split across 10 seeds, matching the sweep size used for Table 10.

Table 13: Test accuracy over 10 seeds on the original and deduplicated splits. Identical configuration and hyperparameters throughout.

	Original	Deduplicated
Test images	1,311	1,092
Mean accuracy	99.22%	99.42%
Standard deviation	0.16 pt	0.30 pt
Range	98.93–99.47	98.81–99.73
Errors	10.2 $\pm$ 2.0	6.3 $\pm$ 3.3

Accuracy on the clean split is 99.42%, which is 0.20 points

higher than on the contaminated split, not lower. A Welch’s t -test on the two sets of ten runs gives $t = 1.89$, $p = 0.081$: not significant at the conventional 0.05 threshold, but closer to it than an earlier five-seed-per-condition version of this comparison suggested ($p = 0.65$), and we report that shift honestly rather than stopping at the version that told a cleaner story. The 95% confidence interval for the deduplicated mean, [99.21%, 99.64%], still contains the original mean, and critically the point estimate still moves in the wrong direction for the leakage-inflation hypothesis: if duplicated test images were propping up the accuracy, removing them should lower it, not raise it. A 95% confidence interval on the difference itself is more informative than the p -value, and we report it as primary: $[-0.03, +0.43]$ percentage points. Because the leakage hypothesis predicts that removing duplicates should *lower* accuracy, the relevant bound is the lower limit, and the data are inconsistent with a duplicate-driven inflation larger than about 0.03 points. An earlier version of this work stated this bound as “roughly a third of a point”, which understated the result by an order of magnitude by reasoning from the p -value rather than from the interval. The same qualitative pattern holds without retraining at all: taking the original seed-42 model, trained with all duplicates present, and simply scoring it on the clean test set gives 99.27%, and on the more aggressive near-duplicate-free test set of 892 images, 99.10%; the model does not depend on the duplicated images it saw during training.

Two things explain this, and together they close the question.

The first is that the duplicated images were never being memorized in the first place. Across all ten models on the released test set, the 216 duplicated test images are misclassified at 1.39% against 0.66% for the other 1,095, a ratio of 2.11. Duplicated images are missed *more* often, not less. Per-image agreement across seeds points the same way: the duplicated images draw a unanimous prediction from all ten models on 92.6% of cases against 97.6% for the rest, so they are also *less* stable run to run. A model coasting on memorized copies would show the opposite on both counts.

We report the size and direction of that effect rather than a significance claim, because the obvious test is not valid here. Treating the 13,110 pooled predictions as independent trials (ten models scored on the same fixed images, which agree with one another on over 98% of cases) understates the standard error substantially; an earlier version of this work did exactly that and reported $p = 0.0011$, which we withdraw. Taking the model as the unit of replication, the mean per-model difference is +0.73 percentage points with a 95% confidence interval of $[-0.22, +1.69]$ (paired t , $p = 0.12$; Wilcoxon $p = 0.23$). Taking the image as the unit, a permutation test that shuffles the duplicate label across images while keeping each image’s ten predictions intact gives $p = 0.060$. Neither reaches conventional significance at this sample size.

The class-stratified view (Table 14) is more informative than either test. Glioma contributes no duplicated test images at all and is the hardest class, at 1.47% error, so it sits entirely in the non-duplicated reference group and drags that group’s error

rate up. The unadjusted comparison is therefore biased *against* the effect we observe. Within each class that actually contains duplicates, duplicated images are missed more often than non-duplicated ones: 1.42% against 0.80% for meningioma, and 1.43% against 0.00% for `notumor`. The argument we need does not depend on a p -value: it is enough that the effect is not negative, and it is not negative in any class that can exhibit it.

Table 14: Error rate on the released test set by class and duplication status, pooled over 10 seeds. Glioma has no duplicated test images, so it contributes only to the reference group.

Class	Duplicated		Not duplicated	
	n	Error	n	Error
glioma	0	—	300	1.47%
meningioma	106	1.42%	200	0.80%
notumor	105	1.43%	300	0.00%
pituitary	5	0.00%	295	0.41%
All	216	1.39%	1,095	0.66%

The second, and the more important, is that deduplication never removed the contamination that matters. Section 3.3 shows that 756 of 756 matched test images in the deduplicated split still belong to a patient present in training, exactly as in the original split, and the more aggressive near-duplicate variant gives the same 100%. Deduplication removes repeated *images* while leaving every repeated *patient* in place. Asking whether it changes accuracy was, in retrospect, asking whether a small subset of redundant images mattered while the far larger source of optimism went untouched. It did not change accuracy because it could not. Section 6.8 runs the experiment that actually removes the leakage.

One caveat on the comparison itself: it is not like-for-like in the strict sense, because deduplication necessarily changes the test set it is measured on, so this is accuracy on a clean test set against accuracy on a contaminated one rather than a paired difference.

An earlier version of this work additionally argued that contamination suppressed run-to-run variance, on the grounds that the original split’s seed-to-seed spread fell below a binomial sampling floor. We have withdrawn that argument. Binomial sampling error describes the uncertainty of an accuracy estimate when the test set is resampled; across seeds the test set is fixed and only the model changes, so there is no such lower bound to fall below, and two independently trained models can legitimately agree on nearly every image. The agreement measurements above are what that claim should have been in the first place, and they point the other way.

Per-class F1 on the clean split, averaged over all ten seeds, is 0.9929 ± 0.0047 for glioma, 0.9845 ± 0.0075 for meningioma, 0.9992 ± 0.0012 for `notumor`, and 0.9971 ± 0.0011 for pituitary. Glioma and meningioma are where the remaining difficulty lives, and notably glioma is the class with no contamination at all, so its difficulty was never an artifact to begin with.

6.8 A Patient-Disjoint Split

Sections 3.2 and 6.7 establish that removing duplicated images changes nothing, and Section 3.3 explains why: every matched test image belongs to a patient the model has already trained on. Recovering the figshare patient identifiers makes it possible to remove that leakage and measure what the pipeline actually does on unseen patients. The value of doing so is established by prior work (Section 2.1); what is specific to this dataset is that the identifiers had to be recovered before the measurement could be made at all.

Construction. We partition the 233 figshare patients into five folds with `build_cv_folds.py`, each patient assigned to exactly one fold, so that across the five folds every patient is predicted exactly once by a model that never trained on them. Patients contribute between 2 and 46 slices, so folds are balanced by greedy longest-processing-time assignment within each class rather than at random; the resulting folds differ by at most two slices per class. Two patients contribute slices to more than one class and are assigned by dominant class. The 437 tumor images with no recovered identifier go to training in every fold and are never eligible for test. The `notumor` class has no patient structure, so we group it by near-duplicate cluster instead: Section 3.4 shows that deduplication alone leaves that class 35% redundant, so a deduplicated-but-random split would still put near-identical scans on both sides. Cluster membership is the closest available proxy for subject identity there, and it is a lower bound on correct grouping rather than a guarantee. Each fold is checked before use for group overlap and for pixel-identical images across the boundary; both are zero by construction and verified.

A defect in our own pipeline. Building this split exposed an error in the work reported above, which we correct here rather than quietly fix. Our training code split train and validation *stratified by class*, not grouped by patient. In every experiment in this paper prior to this section, and in every published result we are aware of on this dataset, the checkpoint, the learning-rate schedule, early stopping, and the temperature fitted in Section 6.6 were therefore selected against a validation set drawn from the same patients as the training set. On our first patient-disjoint fold, 80.3% of validation images belonged to a patient also present in training. Model selection was being performed against exactly the contamination this paper is about. All runs reported in this section use a patient-grouped validation split; the fix is a change of splitter, and the assertion that no group straddles the train/validation boundary now runs on every training job.

Results. We ran three seeds per fold, with seed indices aligned across folds, so that each seed constitutes one complete pass over all 233 patients. Table 15 reports the pooled estimate.

Table 15: Accuracy across the three splits. The first two differ only in whether duplicated images were removed; the third additionally removes patient overlap and is cross-validated over all 233 patients.

Split	Test n	Accuracy
Released	1,311	99.22% $\pm$ 0.16
Deduplicated	1,092	99.42% $\pm$ 0.30
Patient-disjoint (pooled CV)	6,292	95.19%
<i>tumor classes only</i>	4,586	93.68%

Pooled over all five folds, accuracy is 95.19% (replicates 94.91, 95.42, 95.23), against 99.22% on the released split. A patient-level bootstrap, resampling the 233 patients with replacement so that within-patient correlation between slices is respected, gives a 95% interval of roughly [93.5%, 96.7%]. Restricting to the three tumor classes, where the split is fully patient-disjoint and the easy `notumor` class cannot flatter the average, accuracy is 93.68% and macro F1 is 0.935.

Per-class F1 on the tumor classes is 0.9596 ± 0.0029 for pituitary, 0.9446 ± 0.0044 for glioma, and 0.9004 ± 0.0082 for meningioma. Meningioma, which scored 0.987 on the released split, loses nearly nine points and is the class that suffers most. Pituitary tumours sit in a stereotyped anatomical location and transfer to unseen patients comparatively well; the classes whose appearance varies more between individuals do not. None of this is visible on the released split, where all three classes sit above 0.98 and appear equally solved.

A paired comparison. The comparison above comes with the caveat that the two conditions are measured on different test sets. That caveat can be removed entirely. Every released-split test image with a recovered patient identifier appears in exactly one CV fold, so it has a prediction from a model that never saw its patient, and it also has predictions from the ten released-split models that did train on that patient. We restrict this comparison to the 761 tumor test images with a recovered patient identifier, so that the contrast is genuinely between seen and unseen *patients*. A further 300 `notumor` images are scored under both regimes but are excluded, because that class has no patient structure and is grouped only by near-duplicate cluster (Section 3.4); including them would describe 300 images as coming from a seen or unseen patient when they have no patient at all. On those 761 images, with identical images, identical labels, identical architecture and identical hyperparameters, accuracy is 99.12% when the patient was in training and 93.60% when it was not. The error rate is **7.3 times higher** on unseen patients. The only difference between the two conditions is whether the model had seen the patient.

Which uncertainty dominates. Running several seeds on several partitions lets us separate two sources of variation that a single experiment confounds. A one-way random-effects decomposition over the 5×3 table of per-fold accuracies gives a seed component of $\hat{\sigma}_{\text{seed}} = 0.83$ points (10 df) and a partition component of $\hat{\sigma}_{\text{partition}} = 2.18$ points (4 df). Partition choice

contributes 2.6 times the standard deviation that random initialization does. Fold means range from 92.3% to 98.2%, a spread of nearly six points depending on which patients happen to land in the test fold.

This has a direct consequence for how any single-split number on this dataset should be read, including an earlier version of our own. Combining the two components, a single run on a single random patient split carries a standard deviation of about 2.3 points. We previously reported a patient-disjoint result of $94.75\% \pm 0.80$ from one partition; that interval was seed variance with the partition held fixed, which is the smaller of the two terms, and it understated the real uncertainty by roughly a factor of three. We retire that figure in favour of the cross-validated estimate. The same caution applies to every single-split accuracy in Table 16, ours included: at this level, differences of one or two points between methods evaluated on one partition each are not interpretable.

What this does and does not establish. The result is one five-fold partition of one dataset with one architecture, not a claim about brain tumor classification in general, and the bootstrap interval is conditional on the training procedure rather than propagating training variability. The four-class figure mixes a patient-disjoint tumor evaluation with a `notumor` class grouped only by near-duplicate cluster, which is why we report the tumor-only number alongside it. What the result does establish, with a comfortable margin and in a paired design that controls for everything else, is that the near-ceiling accuracy this benchmark is known for depends substantially on evaluating models on patients they were trained on. The honest figure for this pipeline on unseen patients is in the low-to-mid nineties, not 99%.

6.9 Comparison with Prior Work

Table 16 places our result next to recent transfer learning work on brain tumor MRI. We want to be careful about how much weight this carries, and after Sections 3.2 and 6.5 there are three separate reasons for caution. These studies use different dataset versions, different splits, and in one case only three classes. None of them, as far as we can tell from their published descriptions, check for or control for duplicate contamination in the split they evaluate on. And the seed-to-seed spread we measure, 0.16 points on the released split and 0.30 after deduplication, is larger than most of the gaps between the entries in this table, so single-run differences of a few tenths of a point between any of these methods are not interpretable.

The honest reading is that a plain fine-tuned ResNet-18, smaller and cheaper than most of the architectures listed, lands in the same range as heavier models and ensembles, and that this range is where essentially everything lands on a saturated benchmark. That is useful mainly as a floor: if a more complex method cannot beat this baseline by a margin clearly exceeding seed noise, the added complexity is hard to justify.

Table 16: Reported accuracy in recent related work, under each study’s released split. None of these studies report checking their split for duplicated images or for patients shared across the boundary.

Work	Model (dataset)	Acc.
Deepak and Ameer (2019)	GoogLeNet features (figshare, 3-class)	98.00%
Chen et al. (2024)	Multi-CNN feature fusion	>95%
Babu Vimala et al. (2023)	EfficientNetB2 (figshare, 3-class)	99.06%
Disci et al. (2025)	Xception (same release)	98.73%
This work	ResNet-18 (this exact split)	99.22%
This work	ResNet-18, patient-disjoint	see §6.8

One entry deserves particular attention. Disci et al. (2025) report using this release, the same 7,023 images under the same 5,712/1,311 split, and report 98.73%. We did not inspect their code, so we cannot state with certainty that the folders were used exactly as distributed; their reported split sizes indicate that they were. If so, everything established in Section 3 applies to their number as it does to ours: the same 216 duplicated test images, and the same absence of unseen patients among the traceable ones. The two studies by Deepak and Ameer (2019) and Babu Vimala et al. (2023) evaluate on the figshare collection directly, which is the source of the patient structure and contains multiple slices per patient, so a slice-level split there carries the same risk unless patients were explicitly separated, which neither reports doing. We make no claim that any of these results are wrong on their own terms. The point is narrower and applies equally to our own headline number: on a benchmark split this way, none of these figures measure generalization to a new patient, and comparing them to three decimal places is not meaningful.

7 Interpretability

A model that predicts well but cannot explain itself is a hard sell in a clinical setting. We include a Grad-CAM implementation (Selvaraju et al., 2017) that highlights the image regions most responsible for a prediction by backpropagating gradients into the final convolutional stage of the backbone (Layer 4 in Figure 6). Figure 11 shows one example per class, all four correctly classified.

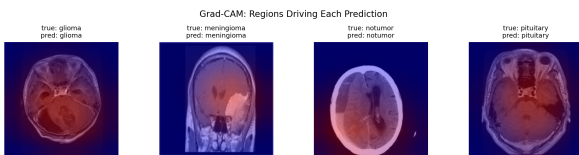


Figure 11: Grad-CAM heatmaps for one example per class. Warmer regions contributed more to the predicted class.

In all four examples attention falls on brain tissue rather than on scan borders, background, or the black region outside the skull. The meningioma example is the most convincing, with the hottest region sitting over the enhancing lesion itself, and the pituitary example concentrates near the skull base where the pituitary sits.

We are wary of reading much into this. It is tempting to present it as ruling out source-shortcutting, the classic failure mode where a dataset aggregates sources with different acquisition conventions and the network learns to recognize the source rather than the pathology. Four hand-picked, correctly classified images cannot establish that. The concern is also specifically live here: this dataset is documented as aggregating three sources, of which we verified the figshare component (Section 3.1), the `notumor` class is 44% redundant against itself (Section 3.4), and the tumor classes arrive at a different orientation from the source collection, all of which are exactly the conditions under which a network can separate classes by provenance cue rather than by anatomy. Establishing that it does not would need a systematic study over the full test set, ideally with the background masked out to force the decision onto tissue. We did not run one. These four maps are an illustration of what the method produces, not evidence about what the model relies on.

That said, we would overstate things by calling these tight localizations. The maps are diffuse, often covering a large fraction of the parenchyma, which is expected given that Grad-CAM at Layer 4 operates on a 7×7 spatial grid upsampled to full resolution. Grad-CAM is a sanity check on what the model is looking at, not a segmentation, and a plausible-looking heatmap is not evidence that a prediction is correct.

8 Limitations and Future Work

The most important caveat is the one the patient-disjoint experiment only partly addresses: even a patient-disjoint split drawn from this collection is still a single-source, same-distribution evaluation. All of the imaging conventions, scanners, and pre-processing are shared between the two sides of the boundary. Removing patient overlap makes the benchmark measure something real, but what it measures is generalization to a new patient within the same collection, not generalization to a new hospital. Multi-institution evaluation remains the single most valuable next step, and nothing in this paper substitutes for it.

A related limit on that experiment is the `notumor` class. It has no recoverable patient identifiers, so it cannot be patient-separated, and our patient-disjoint split handles it by deduplication and random assignment. Any residual patient overlap within `notumor` is therefore unmeasured, which is precisely why Section 6.8 reports the three tumor classes separately: that subset is the number with no asterisk attached. A further 437 tumor images did not match figshare and were assigned to training only, so they cannot contaminate the test set, but they also mean our patient recovery is not exhaustive.

Turning from the split to the measurements made on it, we now have an estimate of run-to-run noise (Section 6.5), and it

is not negligible: test accuracy across ten seeds ranged from 98.93% to 99.47%, a standard deviation of 0.16 points. That means a reported accuracy anywhere in roughly that range from a single run, ours or anyone else’s, is consistent with the same underlying model and training recipe. Table 16’s comparison against other single-run numbers from the literature should be read with this in mind: at this accuracy level, the differences between competing methods are frequently smaller than the seed-to-seed noise of any one of them, and a fair comparison would need a seed sweep on every method being compared, not just this one.

Next, the model is confidently wrong on a meaningful share of its errors, as Section 6 describes, and we followed up on this directly with temperature scaling (Section 6.6). The result was more mixed than we expected: aggregate test-set calibration barely moved, because it was already good, but the confidence of the specific high-confidence errors dropped substantially, cutting the number of errors above 0.90 confidence from four to one. Deeper uncertainty estimation, for example from deep ensembles or Monte Carlo dropout, was out of scope here and remains a reasonable next step, particularly because a single temperature fit on 1,143 validation images is a coarse instrument compared to per-sample uncertainty.

The model also reads a single 2D slice and ignores the volumetric structure of the scan. The dominant glioma-to-meningioma confusion is plausibly a direct consequence: the cues that separate those two often live in tumor location and extent across slices. A 3D CNN or a model that aggregates predictions across neighboring slices is the natural fix and is likely to help precisely where this model is weakest.

The ablation in Section 6.5 averages 5 seeds per cell rather than one, which is enough to say confidently that the 15.65-point gap between full fine-tuning and a frozen backbone is real and that augmentation reverses sign between the two fine-tuning regimes. It is not enough to be fully confident about the smaller full-fine-tuning augmentation effect itself: at 0.56 points with $p = 0.053$, a modestly different seed set could plausibly have landed on the other side of significance. We also did not search over learning rate, batch size, or scheduler settings. A full hyperparameter sweep, repeated across seeds, would make the baseline more informative.

The deduplication comparison in Section 6.7 is a null result, and null results bound effects rather than excluding them. It moved from $p = 0.65$ at five seeds per condition to $p = 0.081$ at ten, always in the direction opposite to the leakage-inflation hypothesis. Since Section 3.3 shows deduplication leaves patient leakage entirely intact, we now regard that comparison as answering a narrower question than it first appeared to: whether repeated images specifically matter, not whether contamination matters.

Finally on method, our provenance work reached the wrong answer before it reached the right one, and the reason generalizes. Content matching between an aggregated dataset and a candidate source is only as good as its negative control. Ours was a pair of identical files, which certified nothing about robustness to the transformations that actually occur, and the resulting false negative survived until we constructed a control

that simulated the release pipeline. Any study attempting this kind of trace should calibrate against transformed matches, not identical ones. We also tested only the figshare component: the SARTAJ and Br35H attributions remain unverified, and 437 tumor images (8.7%) did not match at threshold for reasons we have not diagnosed.

A confound specific to the per-class results deserves naming. Only 58% of meningioma test images could be traced to a patient, against 96% and 99% for glioma and pituitary, and the 437 untraced tumor images are assigned to training in every fold. Meningioma is also the class that degrades most on the patient-disjoint split. We cannot fully separate these: some of that degradation may reflect a distribution difference between the traced and untraced subpopulations rather than the removal of patient leakage. Disambiguating it would require establishing the provenance of the untraced images, which we were not able to do.

Finally, this system only classifies, and it assumes it is handed a usable, roughly centered slice. A real diagnostic tool would need segmentation, automated quality control for artifacts and off-target slices, and integration with patient history to resemble how a radiologist actually works a case.

9 Ethical Considerations

Automated tumor classification raises real ethical questions that should not be glossed over. Tools like this one are meant to support a radiologist’s judgment, not replace it, and this model does make mistakes: it missed two gliomas outright in a test set of 300, and it made several of its errors with high confidence. A false negative can mean a missed diagnosis, and a false positive can mean unnecessary anxiety and follow-up procedures. Any deployment needs a human in the loop and a workflow designed around the assumption that the model will occasionally be confidently wrong.

Training data should be collected with clear patient consent and proper anonymization, and datasets should be as demographically diverse as the populations a model will actually serve. A model validated only on data like the data in this paper should not be presented as validated for general clinical use.

The audit in Section 3 raises a related concern about benchmark hygiene that we think has an ethical dimension in a medical context. A dataset whose splits share 16.5% of their images outright, and whose every traceable test patient also appears in training, is a weak foundation for claims about diagnostic performance. Removing the duplicated images leaves the headline number essentially unchanged; removing the shared patients costs it several points (Section 6.8). The defect that was easiest to see was not the one that mattered. Published accuracies on such a benchmark can acquire an authority they have not earned, and in a clinical setting that authority is exactly what is dangerous. We would encourage anyone building on this dataset to run the audit tooling released here on their own copy before reporting results, and to say plainly which split they used.

We release this code openly so the methodology can be checked, challenged, and improved by others, which we think is a minimum bar for work aimed at clinical settings.

10 Conclusion

A ResNet-18 fine-tuned on this benchmark scores 99.22% on the released split. The same pipeline, trained and evaluated identically but on a split where no patient appears on both sides, scores several points lower, and the gap is not attributable to the duplicated images that are the most visible defect. The released split cannot measure generalization to a new patient, because for the tumor classes it contains no new patients: every test image we could trace belongs to someone the model has already seen.

We want to be careful about the scope of that claim. It is a statement about one dataset, evaluated with one architecture, on one set of cross-validated patient partitions. It is not a claim that the published results built on this benchmark are wrong on their own terms, and it applies to our own headline number exactly as it applies to anyone else's. Nor is the underlying phenomenon novel: slice-level leakage in medical imaging is well documented (Section 2.1). What was specific to this collection is that the patient identifiers had been discarded during aggregation, which made the leakage appear unmeasurable and left it unmeasured.

Recommended protocol for this dataset. Rather than leave this as a complaint, we release what is needed to avoid the problem. `figshare_patient_map.json` gives a recovered patient identifier for 4,586 of the 5,023 tumor images, and `cv_folds.json` gives a fixed five-fold patient-disjoint partition with every patient assigned to exactly one test fold. We suggest that work using this dataset (i) report which split was used, (ii) evaluate on patient-disjoint folds rather than the released folders, and (iii) group the train/validation split by patient as well, since model selection is corrupted by leakage just as evaluation is. Both artifacts are derived metadata rather than redistributed images, so they can be used without reference to the licensing of the underlying collections.

Code and Data Availability

All code, the recovered patient map, the fold assignment, per-run metrics, and per-image predictions are available at <https://github.com/imjbassi/Brain-Tumor-MRI-Classification>. The repository contains the audit tooling (`audit_leakage.py`, `match_figshare.py`, `provenance_sensitivity.py`, `count_unique.py`), the split constructors (`build_clean_split.py`, `build_cv_folds.py`, `build_patient_split.py`), and the training and evaluation pipeline, with pinned dependency versions.

The image data is third-party and is not redistributed here. The aggregated four-class collection is available from Mendeley Data (Ghaffar, 2024) and mirrored on Kaggle (Nickparvar, 2021); the figshare collection used for provenance recovery is Cheng et al. (2015), released under CC BY 4.0. We release only derived metadata (patient identifiers, fold assignments, file manifests) together with scripts that rebuild every split from a local copy of the source data.

Ethics Statement

This work is a secondary analysis of publicly available, de-identified imaging data and did not require institutional review board approval. Consent for the original acquisitions is a matter for the source collections and is described in their respective publications. No patient-identifiable information was accessed, recovered, or released: the “patient identifiers” recovered here are the arbitrary within-dataset labels distributed by the source collection, and they permit grouping slices by subject without revealing anything about who that subject is.

Use of AI Assistance

An AI coding assistant (Anthropic Claude) was used substantially throughout this work: to write and revise analysis code, to run experiments, to draft and edit the manuscript, and to review intermediate results. All experiments were executed on the author's hardware and every reported number is produced by the released scripts and recorded in machine-readable artifacts in the repository. The author verified the analyses, checked the cited literature against the primary sources, and takes full responsibility for the content of this paper, including any remaining errors.

Competing Interests

The author declares no competing interests. This paper discusses limitations of a publicly released dataset and comments on published results that evaluate on it; the author has no affiliation with, or interest in, the parties responsible for either.

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